DIS Department of Mathematics

MYP Grade 10 Extended – Basics of Trigonometry Assessment

Name: Study Guide – Answers Date: ____________________

You will be graded against criteria A (no calculator) and D (GDC required). Unless stated otherwise all final answers are to be given exactly or correct to three significant figures.

Formulas:

Reciprocal trigonometric ratios:

$csc\theta = \frac{1}{sin\theta}$ $sec\theta = \frac{1}{cos\theta}$ $cot\theta = \frac{1}{tan\theta}$

Converting from radians to degrees: $\times \frac{180}{\pi}$

Converting from degrees to radians: $\times \frac{\pi}{180}$

Circumference of a circle: $C = 2\pi r$

Area of a circle: $A = \pi r^2$

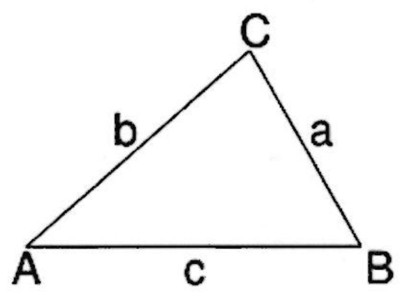
Arc length: $l = \theta r$ (θ in radians) and $l = \frac{\pi r\theta}{180°}$ (θ in degrees)

Sector area: $A = \frac{1}{2}\theta r^2$ (θ in radians) and $A = \frac{\pi r^2\theta}{360°}$ (θ in degrees)

Area of any triangle: $A = \frac{1}{2}absinC$

Sine Rule: $\frac{sinA}{a} = \frac{sinB}{b} = \frac{sinC}{c}$

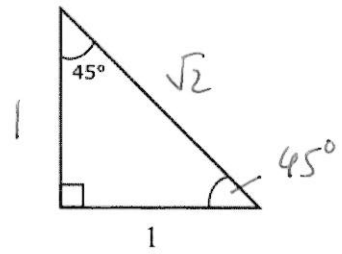
Cosine-Rule: $c^2 = a^2 + b^2 - 2abcosC$

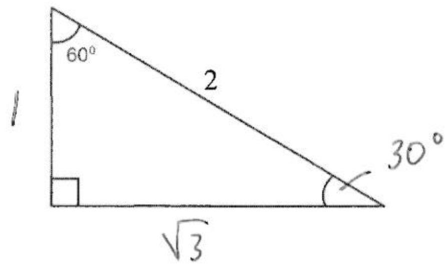


Pythagorean Identity: $sin^2\theta + cos^2\theta = 1$

Criterion A questions (no calculator)

1. Fill in all missing sides and angles in the triangles below.





2. Complete the table below.

Angle in degrees	Angle in radians
45°	$\frac{\pi}{4}$
60°	$\frac{\pi}{3}$
70°	$\frac{7\pi}{18}$
100°	$\frac{5\pi}{9}$
180°	π
$\left(\frac{450}{\pi}\right)^\circ$	2.5
225°	$\frac{5\pi}{4}$
270°	$\frac{3\pi}{2}$

3. Evaluate the following expressions. **Give all answers in simplified form as integers or fractions with positive integer denominators.**

(a) $sin(180°) =$ 0

(b) $tan\left(\frac{\pi}{3}\right) =$ $\sqrt{3}$

(c) $cos\left(\frac{5\pi}{6}\right) =$ $-\frac{\sqrt{3}}{2}$

(d) $tan\left(\frac{7\pi}{4}\right) =$ -1

(e) $tan(4.5\pi) =$> does not exist

(f) $cos(240°) =$ $-\frac{1}{2}$

(g) $csc(0) =$ $\frac{1}{\sin 0}$ => does not exist

(h) $cot(315°)$ = $\frac{1}{\tan(315°)} = \frac{1}{-1} = -1$

(i) $sec\left(\frac{\pi}{2}\right)$ = $\frac{1}{\cos\frac{\pi}{2}} \Rightarrow$ does not exist

(j) $\frac{sin\left(\frac{5\pi}{4}\right)}{cos\left(\frac{7\pi}{6}\right)}$ = $\frac{-\frac{\sqrt{2}}{2}}{-\sqrt{3}/2} = \frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{6}}{3}$

4. Evaluate and write your answer in the form $\frac{1}{a}(\sqrt{a}-1)$, where a is a positive integer:

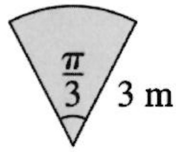
$\cos(240°) + \sin(45°)$

$= -\frac{1}{2} + \frac{\sqrt{2}}{2} = \frac{-1+\sqrt{2}}{2} = \frac{1}{2}(\sqrt{2}-1)$

5. (a) Calculate the length of the arc with angle $\theta = 45°$ and circle diameter 20 cm.

$l = \frac{5\pi}{2}$ cm

(b) Calculate the area and perimeter of the sector shown below.



$\text{Area} = \frac{3\pi}{2} \ m^2$

$P = 2 \cdot 3 + \pi = 6 + \pi \ m$

(c) Given that a sector has radius 2 cm and area π cm², find the sector's angle θ in radians and degrees.

$\pi = \frac{1}{2}\theta \cdot 2^2$

$\rightarrow \quad \theta = \frac{\pi}{2}$

$\theta = 90°$

6. Given that $sin\theta = -\frac{1}{3}$, find the possible values of $cos\theta$. Write your answer in simplest form as a fraction with a positive integer denominator.

$\sin^2\theta + \cos^2\theta = 1$

$\rightarrow \quad \left(-\frac{1}{3}\right)^2 + \cos^2\theta = 1 \rightarrow \cos^2\theta = \frac{8}{9} \rightarrow \cos\theta = \frac{\pm 2\sqrt{2}}{3}$

7. If $cos\theta = -\frac{4}{5}$, find the value of $sin\theta$, given that $tan\theta > 0$.

$\left(-\frac{4}{5}\right)^2 + \sin^2\theta = 1 \rightarrow \sin^2\theta = 1 - \frac{16}{25} = \frac{9}{25}$

$\therefore \sin\theta = \pm\frac{3}{5}$

But $\tan\theta > 0 \quad \therefore \sin\theta = -\frac{3}{5}$

8. If $sinx = -\frac{2}{5}$ and $\pi < x < \frac{3\pi}{2}$, find the value of $tanx$. Give your answer in the form $tanx = \frac{2\sqrt{a}}{a}$, where a is a positive integer.

$\left(\frac{2}{5}\right)^2 + \cos^2 x = 1 \rightarrow \cos^2 x = 1 - \frac{4}{25} = \frac{21}{25}$

$\cos x = \frac{\pm\sqrt{21}}{5} \rightarrow \pi < x < \frac{3\pi}{2}$

$\therefore \cos x = \frac{-\sqrt{21}}{5}$

$\therefore \tan x = \frac{-2/5}{-\frac{\sqrt{21}}{5}} = \underline{\underline{\frac{2\sqrt{21}}{21}}}$

9. By first expanding the brackets, simplify $(sinx + 2cosx)^2 + (2sinx - cosx)^2$ to a positive integer number.

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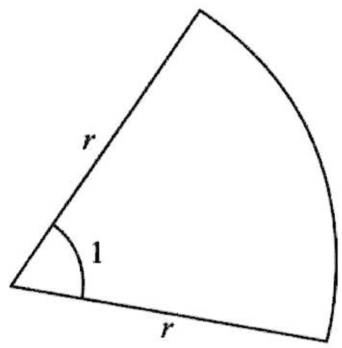
10. A sector of a circle with radius r cm, where $r > 0$, is shown on the following diagram. The sector has an angle of 1 radian at the centre.



Let the area of the sector be A cm^2 and the perimeter be P cm. Given that $A = P$, find the value of r.

$3r = \frac{1}{2} \cdot 1 \cdot r^2$

$6r = r^2 \rightarrow r^2 - 6r = 0$

$r(r-6) = 0$

$\therefore \underline{\underline{r = 6}}$

11. The diagram below shows two circles with centre O and radii 2 cm and 4 cm. The points P and Q lie on the larger circle and angle POQ $= x$, where $0 < x < \frac{\pi}{2}$.

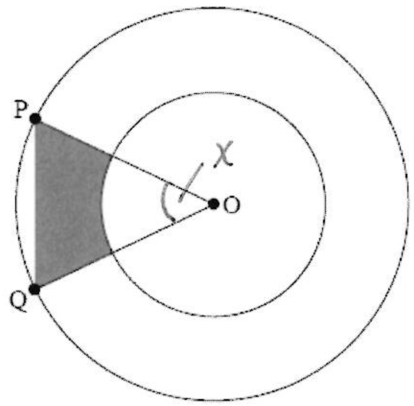


diagram not to scale

Find an expression, in terms of x, that calculates the area of the shaded region.

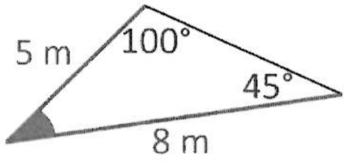
$$\text{Area} = \frac{1}{2} \cdot 4^2 \sin x - \frac{1}{2} x \cdot 2^2$$

$$= \underline{\underline{8 \sin x - 2x}}$$

Criterion D questions (GDC required)

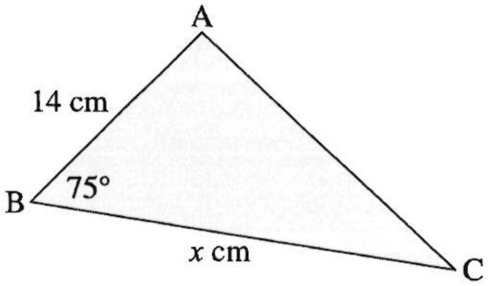
Unless stated otherwise in the question, all answers are to be given correct to three significant figures.

1. Calculate the area of the triangle below.



11.5 m^2

2. The triangle below has an area of 162 cm².

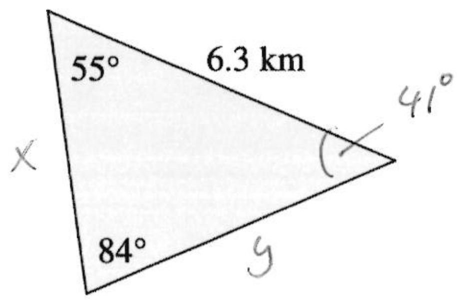


Find the value of x and hence, calculate the perimeter of the triangle.

$x = 24.0$ cm

$\therefore P = 62.5$ cm

3. Consider the triangle below.



(a) Write down the value of the missing angle in radians and degrees.

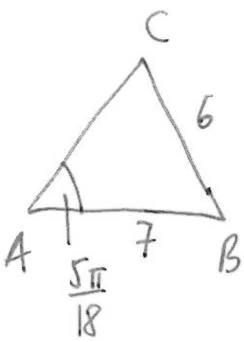
$41° =$

(b) Calculate the values of the two missing side lengths.

$x = 4.16$ km

$y = 5.19$ km

4. A triangle ABC has angle A = $\frac{5\pi}{18}$, AB = 7 cm, and BC = 6 cm. Find the area of the triangle given that it is smaller than 10 cm^2. Round your answer to the nearest cm^2.



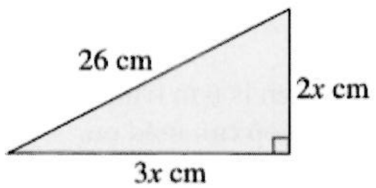
$$\frac{6}{\sin\left(\frac{5\pi}{18}\right)} = \frac{7}{\sin C} \rightarrow C = 1.11 \text{ or } 2.04$$

$$\text{Area} = \frac{1}{2}\cdot 6 \cdot 7 \cdot \sin\left(\pi - \frac{5\pi}{18} - 1.11\right)$$
$$= \underline{19.2\ cm^2} \leftarrow \underline{\text{reject}}$$

or

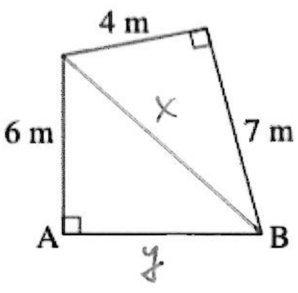
$$\text{Area} = \frac{1}{2}\cdot 6 \cdot 7 \cdot \sin\left(\pi - \frac{5\pi}{18} - 2.04\right)$$
$$= 4.77\ cm^2$$
$$= \underline{\underline{5\ cm^2}}$$

5. Calculate "x" and hence, find the perimeter of the triangle:



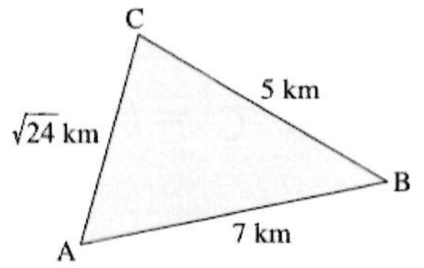
$$(2x)^2 + (3x)^2 = 26^2$$
$$\therefore x = 2\sqrt{13} = 7.21$$
$$\therefore \text{Perimeter} = 3\cdot 7.21 + 2\cdot 7.21 + 26$$
$$= \underline{\underline{62.1}}\ cm$$

6. Find the distance AB in the following figure:



$$x^2 = 4^2 + 7^2 \rightarrow x = \sqrt{65} = 8.06\ m$$
$$\therefore y^2 + 6^2 = 65$$
$$\rightarrow y = \sqrt{29} \approx \underline{\underline{5.39\ m}}$$

7. The following triangle is not drawn to scale. Is it a right triangle? If so, where is the right angle? Justify your answer.

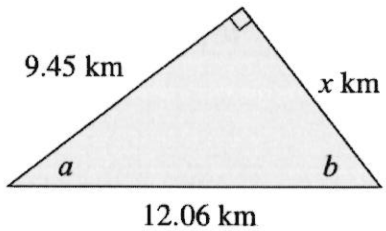


Test: $5^2 + 7^2 = 24$ not true

$5^2 + 24 = 7^2$ true

∴ Its a right triangle and the right angle is at C.

8. Calculate the values of a, b, and x in the figure below:

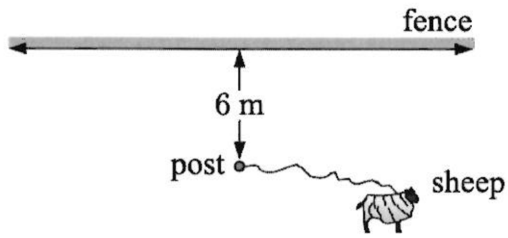


$x = 7.49$

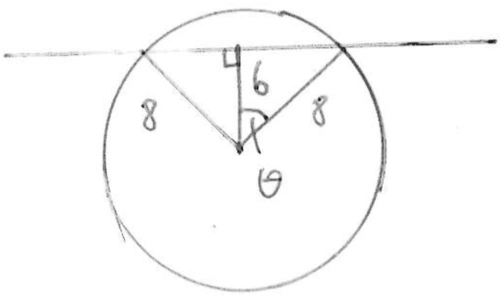
$b = 51.6°$

$a = 38.4°$

9. A sheep stands on a grassy field and is tethered to a post which is 6 m from a long fence. The length of the rope is 8 m. Find the area that the sheep can feed on.



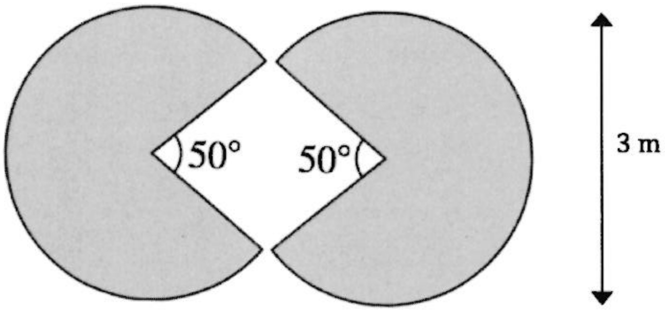
Area = 187 m^2



$\cos\theta = \frac{6}{8} \rightarrow \theta = 41.4°$

$2\theta = 82.8°$

10. A company wants to have their logo painted on the outside of their office building. The logo consists of two congruent sectors with measurements indicated in the diagram below.



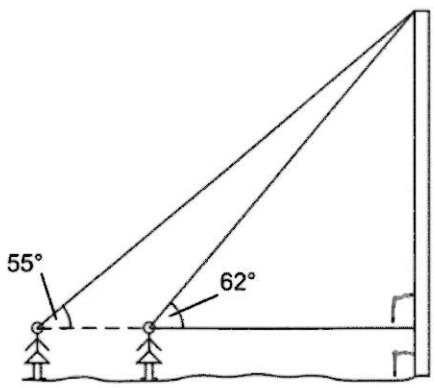
Determine the cost of having the logo painted if the price is 168.23 € per m².

$$\text{Area} = \frac{2\pi \cdot 1.5^2 \cdot 310}{360} = \frac{31\pi}{8}\ m^2 \quad (\approx 12.2\ m^2)$$

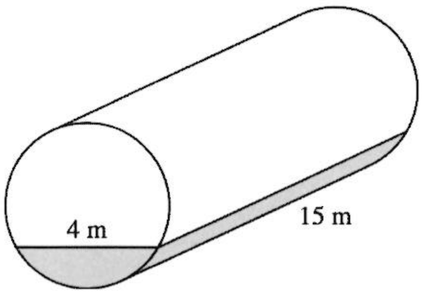
$$\therefore \text{cost} = 12.2 \cdot 168.23 = \underline{\underline{2050\ €}}$$

11. Twins Tansu and Zeynep are both 1.65 m tall. They both look at the top of Frauenkirche (see diagram with angles of elevation below). If the two girls are standing 16 m apart, what is the Frauenkirche's height?

$\underline{\underline{96.6\ m}}$

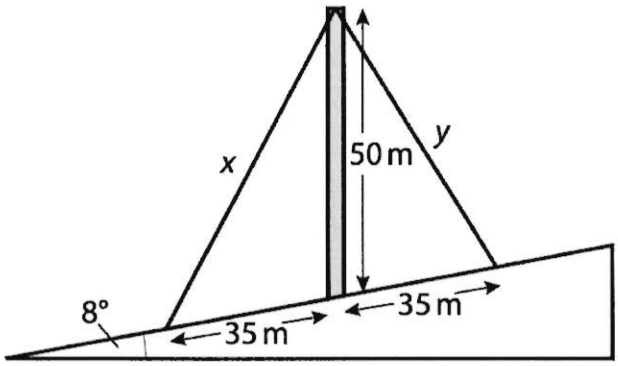


12. Find, in m^3, the volume of water lying in this cylindrical pipe of radius 2.5 m.

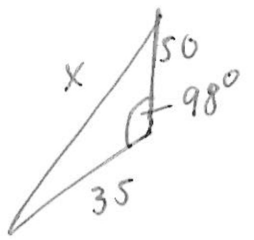


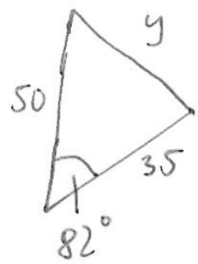
$41.9\ m^3$

13. A 50 m pole is to be placed vertically on the side of a hill that makes an 8° angle with the horizontal (see diagram below).



Find the lengths x and y of the two supporting rods that will be anchored 35 m uphill and downhill from the base of the pole.





$x = 64.9\ m$

$y = 56.9\ m$